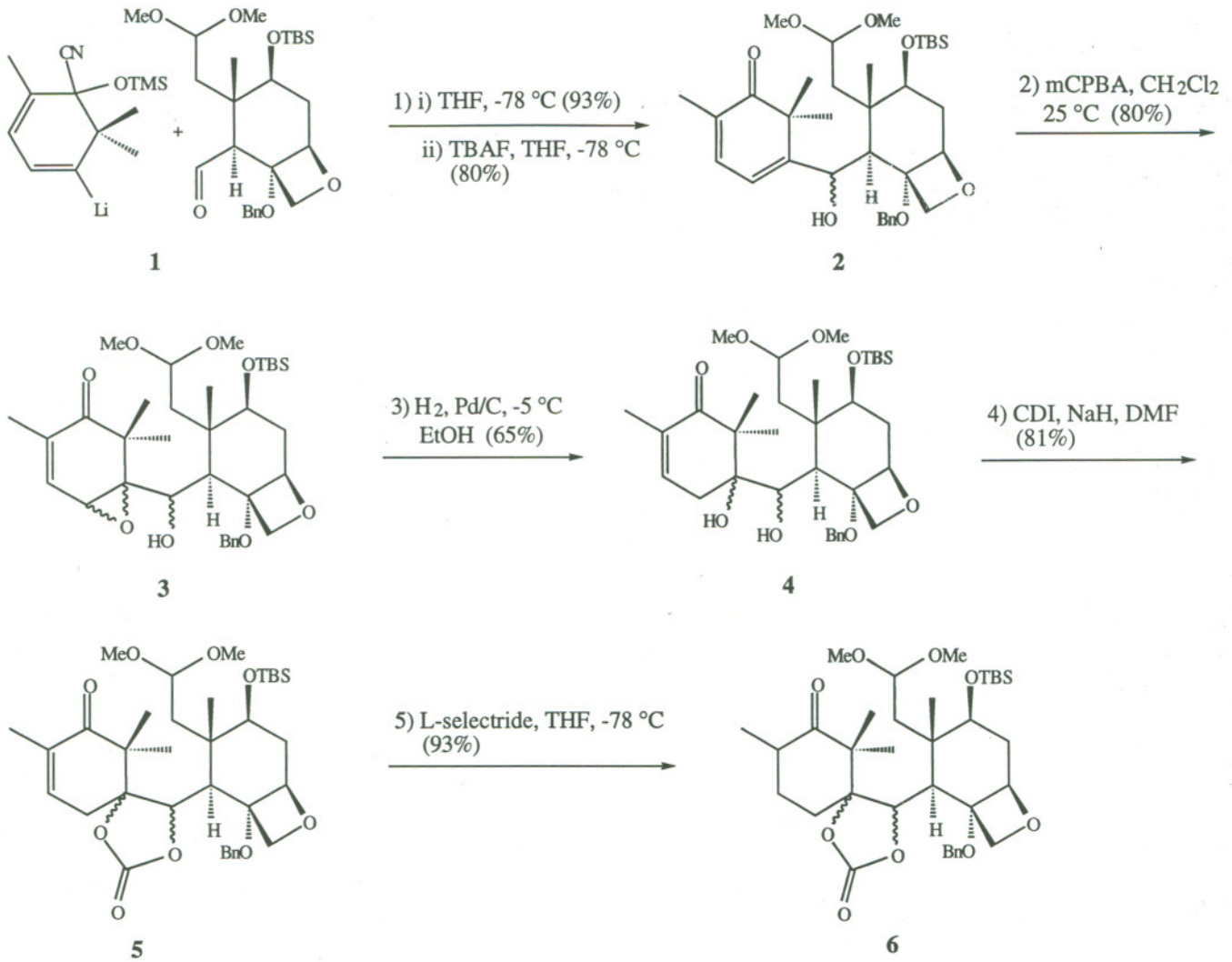
Retake of March 2 Cume

**Organic Cumulative Exam** **April 13, 1996**

A recent article in *J. Am. Chem. Soc.* (Danishefsky, S.J., *et. al.*, *JACS* **1996**, *118*, 2843-59) reported on the "Total Synthesis of Baccatin III and Taxol". Answer the following questions based on the 5 step synthetic sequence shown below which was taken from this paper.




TBS = *t*butyldimethylsilyl; TMS = trimethylsilyl; Bn = benzyl; TBAF = tetrabutylammoniumfluoride; mCPBA = *meta*-chloroperbenzoic acid; CDI = carbonyldiimidazole; L-selectride = Lithium tri-*sec*-butylborohydride

1) The formation of **2** was reported to give a single diastereomer. Show the correct stereochemistry for **2** and rationalize with an appropriate principle for stereoselectivity in acyl addition reactions. Show an appropriate diagram to accompany your rationale. [Hint: Carefully consider the steric differences of the controlling substituents] {Extra Credit given for the name of the principle}

2) Show the expected stereochemistry in the formation of epoxide **3**, based on the stereochemical assignment of alcohol **2** from question #1 (*i.e.*, assign all undesignated stereocenters of **3**). Explain the directing force for the selectivity **with** an accompanying diagram. [Hint: Consider the nature of the particular reagent being used]

3) Explain the regioselectivity of the epoxidation reaction to form **3**. [Hint: Again, consider the nature of the reagent chosen for this reaction]

4) Offer an explanation for the chemo- **and** regioselectivity in the hydrogenolysis reaction to form **4**.

5) Explain the chemoselectivity in the reduction reaction to form **6**. [Hint: Again, consider the nature of the reagent chosen for this reaction]